

LSB  $\rightarrow$  is the rightmost bit

DAY: \_\_\_\_\_

DATE: \_\_\_\_\_

1's complement:

inverts all bits of a binary number.

2's complement:

inverts all bits and add 1 to the LSB.

$\Rightarrow$  1's complement + 1

$(10101)_2 \rightarrow 01010$   
(1's complement)

$1 \rightarrow 0$

$0 \rightarrow 1$

1+1=0  
0+0=0  
1+0=1  
0+1=1

$(10101) \rightarrow 01010$   
 $\begin{array}{r} +1 \\ \hline 01011 \end{array}$

01010

2's  $\rightarrow 10110$   
complement.

10101  
2's complement 01011

01010  
 $\xrightarrow{\text{Psum}}$   
 $\begin{array}{r} 10101 \\ +1 \\ \hline 10110 \end{array}$  ✓



DAY: \_\_\_\_\_

## Practic Questions

$$(81)_{10} \rightarrow B, O, H$$

$$(49)_{10} \rightarrow B, O, H$$

$$(685)_{10} \rightarrow B, O, H$$

$$(25.050)_{10} \rightarrow B, O$$

$$(10011101)_2 \rightarrow D$$

$$(27.4)_8 \rightarrow D$$

$$(FAB)_{16} \rightarrow D$$

$$(428.964)_{16} \rightarrow D$$

$$(43.5)_8 \rightarrow B$$

$$(0100.1)_B \rightarrow O$$

$$(9AD)_{16} \rightarrow O$$

$$(627)_8 \rightarrow H$$